

A pipeline for facial animations on low budget VR productions

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ABSTRACT

This work presents a new approach to developing character facial animations for Virtual Reality projects focusing on low budget productions. Facial animations are mostly time-consuming and complex and often involve expensive equipment and software inaccessible to the vast majority of professionals. This work aims to present an alternative facial animation pipeline using a hybrid approach and combining traditional software, algorithms, and processes to offer alternatives for producing high-quality facial animations for characters focusing on Virtual Reality applications.

Keywords: facial animation, pipeline, motion capture, virtual reality.

Index Terms: Human-centered computing, Human-computer interaction (HCI), Interaction paradigms, Virtual reality

1 INTRODUCTION

Facial animations are mostly complicated and time-consuming. In some cases, they involve expensive equipment and software inaccessible to the vast majority of professionals. Although the idea of using motion capture is attractive and the impacts on production time and quality are undeniable, the high cost of the technology can make its adoption impossible [1]. This reality is even more complicated for small businesses and independent professionals who do not have a large budget and are part of the indie industry segment [2]. The alternative to the high cost of motion capture is the adoption of traditional animation methods. However, such methods require dedicated professionals and have a high production time, usually better in stylized productions [3].

When there is a demand for the realism of facial reactions, it becomes challenging to obtain good results with animations made by traditional methods. Recently many experimental works related to facial animation methods have been produced. Many of them have a starting point in using technologies like machine learning and computer vision [4]–[6].

However, few works focus on viable solutions applied to low-budget productions, especially those dedicated to real-time applications, like Virtual Reality content. However, there are works dedicated to developing hyper-realistic faces driven by cameras on the VR headset [7]. Therefore, this work aims to present a proposal for a facial animation pipeline using a hybrid approach. This proposal combines accessible software, hardware, algorithms, and traditional processes to offer alternatives for producing high-quality facial animations for characters focusing on Virtual Reality applications in low budget productions. This work is organized as follows: after this brief introduction, a small reference of the main concepts covered in this paper is provided. Next, the pipeline's

motivation and presentation will be provided, immediately followed by a description of the components on which it is based. Finally, the results obtained will be addressed, as well as proposals for improvement and limitations.

2 THEORETICAL BACKGROUND

2.1 Facial Animations

To promote a bond between the player and the characters, modern games require a high degree of realism. Studies reveal a strong relationship between high levels of realism and greater engagement from users [8]. However, communication between the player and the game depends on how the virtual character expresses himself, which involves how he moves, acts, and speaks. No doubt, titles with characters that have realistic facial expressions have become a constant. However, the vast majority of these titles come from companies that have a considerable budget, given that the animation activity requires, in addition to a highly specialized workforce, hardware, and software of high cost.

Facial animations present a series of challenges. Some of them are still in full development, as is the case of lip-syncing. In recent years many approaches to facial animations have been proposed.

The two main approaches to lip-sync animation are procedural and data-oriented [9]. The data-driven approaches find a balance between naturalness versus animation control but depend on data collection and processing and expensive and time-consuming processes. Procedural approaches are a better choice in terms of animation control. However, they may not reach the level of naturalness in data-driven approaches [9]. For this reason, this work adopts a hybrid approach as a way to improve the final results.

2.2 Virtual Reality

Virtual reality (VR) is a user interface that provides an immersive computer simulation of a three-dimensional environment. During the VR experience, the person interacts and experiences multi-sensorial experiences from tracking the head's movement. The VR experience comprises three essential elements: immersion, interaction, and engagement [10].

Immersion is the sensation of being totally within an environment. Interaction is how the individual reacts according to the VR stimulus presented. Engagement is the degree of user involvement with the stimulus. These features make VR stand out from other technologies where users passively watch a two-dimensional video. Currently, the evolution of technology, the greater availability of content, and the reduction of costs of VR devices have allowed the popularization and expansion of its use for entertainment, work situations, health applications, and several other areas [11].

Far beyond a simple technology, Virtual Reality is also a powerful medium for transposing reality, making it ideal for various applications. However, with the advancement of technology and game consoles, there has also been an increasing demand for quality content. This demand often impacts the limitations familiar to small studios and independent developers. Besides, some devices still have limitations, especially mobile ones. Those limitations make the production of content even more restricted to a series of constraints such as the number of polygons and the number and size of textures [12].

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2.3 Low budget productions

The phenomenon of end-user-driven technological practices such as the possibility of creating animations and games without the dependency on billionaire budgets brings with it a series of new opportunities and challenges. The availability of high-quality development tools and, above all, new forms of distribution such as marketplaces have attracted more and more independent professionals and small studios. Simultaneously, there is growing consumer demand for high-quality animation, which raises the debate between low cost and high-quality animations. Therefore, generating accessible solutions without giving up quality is a challenge that can bring competitive differential to small studios and independent professionals.

In many industries, the value chain concept is a variable that cannot be ignored. According to Porter [13], every company comprises activities to design, produce, deliver, bring to market and support the product. A value can represent all these activities. Nevertheless, even with the democratization of specific technologies and the change of several paradigms, it is still difficult to compete with the big producers when it comes to some aspects. One of these aspects is the production of content, in some cases, one of the most expensive and complicated parts of game development.

One constant in the independent gaming industry is the question of budget. Being independent has its advantages, such as creative freedom and content definition, according to the author. Lipkin [14] defines the term indie beyond independence. It is quickly co-opted and marketed as a highly marketable genre in the gaming industry. Independent is that which is not mainstream.

Furthermore, the mainstream goal of popularity and profit overcomes creativity, self-expression, and art. Although aesthetics and discourse lead us to believe that this is a stable industry, it is evident that one of the most significant competitive limitations of this segment is the financial one [15]. Budget limitations constitute a significant barrier to competitiveness in some segments, as in virtual reality and AAA games.

Therefore, it makes sense to invest in applied research to take advantage of low-cost technologies to raise production quality without necessarily raising production costs with equipment and software, increasing perceived value.

3 THE PROPOSED FACIAL ANIMATION PIPELINE

3.1 Background and motivation

This paper describes a small part of a more extensive and complex ongoing research (a Ph.D. research) that involves the development of Virtual Reality simulators applied to security professionals and law enforcement agents' training.

The content of this simulator provides, among other things, conflict situations where the reactions of professionals in training are required. Some of the characteristics and technical and aesthetic choices defined during the simulator's construction have resulted in developing convincing and realistic facial characters and animations. In turn, these animations were proposed to stimulate not only a higher level of immersion in the simulator but also to stimulate physiological reactions by the professionals in training. However, as academic research and without funds for acquiring professional motion capture equipment, the challenges for developing animations with the intended quality level have become substantial. As part of the solution, a series of combinations of low-cost software, popular hardware, and techniques were experimented with, which allowed the achievement of the intended results, with some caveats pointed out as limitations.

3.2 Pipeline presentation

The expression "pipeline" is commonly used for animation production processes. It refers to the path that the production

follows to be realized and completed through the different departments involved. Figure 1 presents an overview of the animation pipeline presented in this work with its main components and processes.

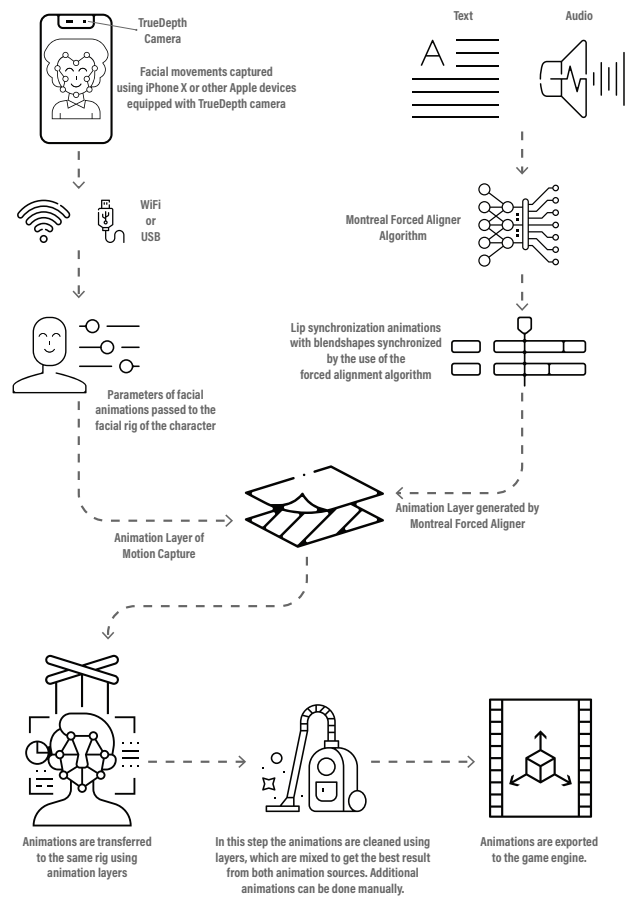


Figure 1: The facial animation pipeline for VR and its interactions.

In this work's specific case, the pipeline presented is dedicated exclusively to a part of the content production process focusing on real-time applications. More specifically, facial animations for Virtual Reality applications. Therefore, it starts after some processes have already been performed, among them, the three-dimensional modeling and facial rigging process. It also presupposes the existence of Blend Shapes with phonemes and facial expressions. Blend Shape is the implementation of the technique that allows the deformation of a single mesh to reach numerous predefined shapes and any number of combinations of these shapes. Since these are real-time applications and their content will be exported in the end to a game engine, we chose to use rigging based on Blend Shapes over other perhaps more efficient methods. There is a discussion about each of the most used rigging and facial animation methods in [16]. Also, regarding Blend Shapes, this work considers the use of FACS (Facial Action Coding System) for the Blend Shapes. This approach simulates a set of facial muscle movements that correspond to a displayed emotion [17].

The pipeline is composed, as already mentioned, by a combination of software, hardware, and techniques having as the main parameter the accessible cost. Therefore, it is convenient to present the main components and their justification to achieve the final goal, more

specifically, facial animation itself in a game engine. Below, we present each of the components divided and classified, more specifically, (1) Hardware, (2) Software, and (3) Technique.

3.2.1 Hardware

As already mentioned, motion capture technologies have become increasingly popular in recent years. However, specific hardware for motion capture is still expensive and not affordable for most small studios and independent professionals. That said, the first step was to try alternative hardware solutions. When this research was at the point of deciding which technologies to adopt, many alternatives were raised. Among those that represented the best cost-benefit was adopting mobile devices equipped with cameras capable of capturing depth information and transferring this information to a character prepared for animation. Starting with the work of Jie et al. [18] that used precisely the same approach to create the facial animations of an animated film successfully, it was decided to adopt a similar workflow, but with some differences, since our case required that the content be published in a real-time application.

The idea is to use depth sensors present in devices such as the iPhone to capture facial expressions and move and transition Blend Shapes. Although the idea is relatively simple and plausible, its implementation depends on the software that bridges the hardware capture to the output itself in real-time. In this case, the connection can be done via USB or Wi-Fi or previously recorded in a format capable of storing motion information, such as the FBX format. This file containing the animations would then be imported into a three-dimensional content production software. The animations would be transferred to the rigging, which would then be exported to the final application.

Regarding the accuracy of the iPhone cameras for motion capture, there is the work of Breitharth et al. [19], which states that the iPhone X TrueDepth system can be used for measuring tasks with millimeter accuracy, which is ideal for facial animations. The principle behind TrueDepth¹ is to use a projector to throw 30,000 points on the face, which are then read by an infrared camera [20], [21], similar to the technology behind Microsoft Kinect [22]. In both technologies, all data is passed to a machine learning system that has been trained with thousands of examples. In this work, the technology's adoption proved to be very convenient, given that the devices that rely on this type of technology are trendy and much more accessible than equipment dedicated to motion capture.

3.2.2 Software

There is a series of software fulfilling specific tasks in the proposed pipeline and used at different moments. First, like all content production, a three-dimensional content creation package was adopted, which in the specific context of this work is the Autodesk Maya [23]. Although other software could replace Autodesk Maya without any negative impact on the pipeline, the option was based on personal experience, production speed, and possibly less time to solve technical problems. When it comes to costs, the software can be considered expensive. However, it has a version focused on the game market and affordable prices, although without the same number of features as the full version, which does not impact anything, considering that the main animation functions are

present. An excellent alternative and free is the Blender software, which has become very popular.

The second software adopted in this research was the motion capture software MocapX [24]. This iPhone application captures facial animations and is capable of transferring the animations to the 3D software. Nowadays, there are several applications with the same purpose, with even more affordable prices and costing only a few dollars, like Face Cap² or more expensive and with integration with specialized hardware such as Rokoko Studio for Facial Motion Capture³. However, the choice was also based on agility and convenience.

While the other applications perform only the part of motion capture using precisely the same technology, MocapX offers a convenient way to connect the mobile device straight to the computer and Autodesk Maya, which makes it possible to stream data directly from the cell phone or, if there is a problem with the network, adopt a workflow where the animations are recorded along with the audio on the mobile device and then transferred to the computer for editing. The only drawback that should be pointed out is the time needed to connect the facial rig to the controllers responsible for replicating the captured data. Still, the application has good documentation. The workflow is very intuitive but requires some knowledge about the facial rigging process.

3.2.3 Technique

After the motion capture and once all the animations are transferred to the respective character, the animations' treatment and correction phase begin. Although there is high precision in motion capture, which directly impacts the generated animations [25], not all animations can be used directly on the final product without some kind of treatment. That is why we adopt a workflow that involves a particular manual effort, obviously much less and much more precise than starting the animations from scratch. One of the most common problems in motion capture refers to the amount of noise and, in some cases, capture inaccuracies and failures, depending on several capture conditions [26], [1].

In the case of this work and because it is an exploration of possible workflows, another approach was chosen: to recreate animations using a combination of text and audio and an alignment algorithm known as Montreal Forced Aligner (MFA) [27], a solution currently adopted by many software and plugins dedicated to lip-sync in animations. MFA adopts a technique that implements a Hidden Markov Model-Gaussian Mixture Model (HMM-GMM) technique to perform the acoustic model training and subsequent alignment [28]. The algorithm works with two input data sources: text and audio. It then uses audio and text to align the phonemes' positions to the preset Blend Shapes and synchronize the animations with the timeline's audio.

The MFA uses the Kaldi ASR toolkit [29] to perform forced alignment, is open source, and can be integrated with Autodesk Maya or any other 3D computer graphics software that supports the integration of third-party software, something that happens with most software on the market. The software has pre-trained models in more than twenty-two languages with accuracy greater than 90%. However, its implementation and use combined with Autodesk Maya is not very intuitive and has required a great deal of development time, even if the software has vast and rich documentation.

¹ Compatible with any iPhone/iPad enabled by TrueDepth camera for facial recognition: iPhone X, iPhone XS, iPhone XS Max, iPhone XR, iPhone 11, iPhone 11 Pro, iPhone 11 Pro Max, iPad Pro 11-in. (1st and 2nd generation) and iPad Pro 12.9-in. (3rd and 4th generation).

² <https://www.bannaflak.com/face-cap/index.html>

³ <https://www.rokoko.com/studio/face-capture>

The MFA is free software made in Python, available for all platforms, and runs all processes in previously trained language bases. To work, it only requires passing the parameters. In the context of this research, the implementation was performed in MEL (Maya Embedded Language), a scripting language used to simplify tasks in Autodesk's Maya. Most of the tasks that can be performed through Maya's interface can be performed with MEL and specific tasks that are not available in the interface, such as integration with third-party software. A simple GUI has been created, which can be seen in Figure 2.

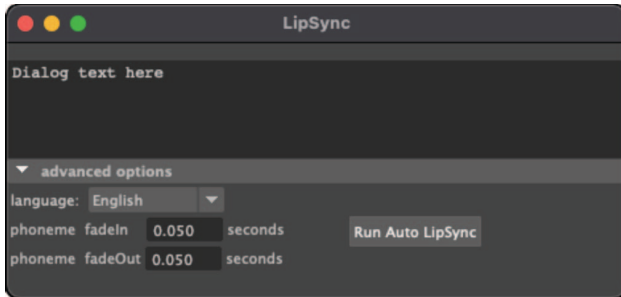


Figure 2: Simple GUI for lip-sync using Montreal Forced Aligner.

This interface receives as input the text of the dialog, the definition of the language, and the description of each phoneme's input and output speed. Furthermore, it can use the audio connected to the timeline to synchronize phonemes with the Blend Shapes, generating the lip-synchronization animations.

At the end of this process, there are two different types of animation for the same character. The next step is based on a traditional method of animation. This process involves separating the two animation sources into layers and working the mixture of these layers, which requires human intervention and judgment to assess each of the animation sources' quality for each specific expression or phoneme. This is done using animation layers, something familiar in most 3D animation software used on a large scale. In Autodesk Maya, animation layers are tools that enable a non-destructive workflow and make it possible to work with different animation weights and intensity. This makes it possible, among other things, to choose to emphasize parts of the animation where the best quality comes from capturing movements and parts where the best quality comes from forced alignment animation.

The main reason to adopt a hybrid and possibly more complex approach is mainly due to the quality of lip synchronicity obtained by using the Montreal Forced Aligner, despite the algorithm's limitation to generate only lip-sync animations. On the other hand, the motion capture's facial animations are better at capturing facial expressions.

At this point, it should be made explicit that only qualitative evaluations of the final result were conducted. These evaluations were based on generating the same dialog of a few seconds using the same character and facial rigidity and generating three animations: one only with the lip synchronization (coming from the use of the forced alignment algorithm), another only with the facial motion capture and finally, animation with the mixture of both. After that, the animations were presented to the appreciation and judgment of professionals with some knowledge about animation, which does not configure a scientifically rigorous evaluation method.

The animations generated by the process of cleaning and mixing animations generated exciting results, but not always superior to

the animations coming from the motion capture, which denotes the quality of facial motion capture and explains, in part, its adoption by the industry in recent years. One of the significant disadvantages of adopting this workflow is the slowness of this final phase (cleaning and treating the animations). However, the slowness is due to human intervention, something characteristic of the animation process and, to some extent, a common practice in the industry [30]. Even if there is high precision in the capture of movement, there will always be the need to change something after the capture is made, whether by aesthetic or narrative decisions. However, eliminating manual work can be a possibility if it is feasible to implement a workflow that can combine approaches such as facial motion capture and machine learning [31], [32]. The final step is to export the animations for a game engine and then proceed with the production.

4 CONCLUSION

This work reports part of ongoing research and focuses on developing a facial animation pipeline that uses a combination of low-cost hardware and software and traditional animation techniques. Although the presented pipeline was developed for a specific Virtual Reality application, it can be adopted without significant changes in virtually any type of production involving facial character animation. However, small studios and independent professionals can benefit greatly from experience reported in this work, especially as it is an effort to propose low-cost alternatives in both hardware and software. There are limitations, and they need to be made explicit about proposing new challenges and advances. The first one refers to the use of some software that, although low cost, have costs. The software used in this research to perform motion capture was not chosen because it was the cheapest (and it was not). In this research, convenience and flexibility were the main reasons since it offers, in addition to the application, a plugin that facilitates the capture process and connects the mobile device to the 3D software.

Furthermore, the forced alignment algorithm's implementation is not simple and required a great effort for deployment. However, there is extensive documentation and different ways to use the API. Finally, it is essential to emphasize that this exploratory work aims to report a workflow that served a particular purpose and scenario. Therefore, it needs more testing and validation. However, even with several technical limitations and dependencies, it is imagined that its application can be extended and tried out in different types of scenarios and applications that go beyond Virtual Reality. Therefore, as a suggestion for future work, it is recommended to apply it to pipelines in scenarios where demand is even higher, such as low-budget animation films. Considering that there are no limitations on form or content, the same pipeline could also be adopted for stylized characters, as long as the same requirements explained in this research are met. This type of scenario commonly requires interactions between characters, which may be another way of testing the pipeline by simultaneously capturing several characters. Another suggestion for future explorations refers to using the forced alignment algorithm in real-time situations, such as avatar animations for immersive experiments.

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⁴ <https://pie-lab.at/>

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